over the old data, in looking with *Viking-type* instruments at Antarctic and other soils that have few microbes in them, in laboratory simulation of the role of oxidants in the Martian soil, and in designing experiments to elucidate these matters—not excluding further searches for life—with future Mars landers.

If indeed no unambiguous signatures of life were determined by a variety of sensitive experiments at two sites 5,000 kilometers apart on a planet marked by global wind transport of fine particles, this is at least suggestive that Mars may be, today at least, a lifeless planet. But if Mars *is* lifeless, we have two planets, of virtually identical age and early conditions, evolving next door to one another in the same solar system: Life evolves and proliferates on one, but not the other. Why?

Perhaps the chemical or fossil remains of early Martian life can still be found—subsurface, safely protected from the ultraviolet radiation and its oxidation products that today fry the surface. Perhaps in a rock face exposed by a landslide, or in the banks of an ancient river valley or dry lake bed, or in the polar, laminated terrain, key evidence for life on another planet is waiting.

Despite its absence on the surface of Mars, the planet's two moons, Phobos and Deimos, seem to be rich in complex organic matter dating back to the early history of the Solar System. The Soviet *Phobos 2* spacecraft found evidence of water vapor being outgassed from Phobos, as if it has an icy interior heated by radioactivity. The moons of Mars may have long ago been captured from somewhere in the outer Solar System; conceivably, they are among the nearest available examples of unaltered stuff from the earliest days of the Solar System. Phobos and Deimos are very small, each roughly 10 kilometers across; the gravity they exert is nearly negligible. So it's comparatively easy to rendezvous with them, land on them, examine them, use them as a base of operations to study Mars, and then go home.

Mars calls, a storehouse of scientific information—important in its own right but also for the light it casts on the environment of our own planet. There are mysteries waiting to be resolved about the interior of Mars and its mode of origin, the nature of volcanos on a